Online Appendix to "The Impact of Central Bank Stock Purchases: Evidence from Discontinuities in Policy Rules"

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A Empirical Appendix

A.1 Local Nonlinear Impulse Response Function

In this section, we allow nonlinearity in the impulse response and show that our estimands can be interpreted as dynamic "local average treatment effects" in the spirit of Angrist and Imbens (1995).

We first define a potential outcome framework in our context following Rambachan and Shephard (2021). For each $t \geq 1$, the BoJ decides the amount of ETF purchases ETF_t; let us denote $ETF_{1:t} \equiv (ETF_1, \cdots, ETF_t)$. Let $w_{1:t} \equiv (w_1, \cdots, w_t)$ be a potential assignment path up to t where $w_t \in [0, \bar{w}]$ for all t. Associated with this potential assignment path, the potential outcome on day $t+l$ at time h is $Y_{t+l,h}(w_{1:t+l})$.[1] Note that for any different assignment paths, there exist different outcome paths but we only observe $Y_{t+l,h}(ETF_{1:t+l})$. For any day $t+l$ and time h, let us denote

$$Y_{t+l,h}(w) \equiv Y_{t+l,h}(ETF_{1:t-1}, \underbrace{w}_{t-th}, ETF_{t+1:t+l}).$$

Using this notation, the observed outcome can be denoted as $Y_{t+l,h}(ETF_t)$ by definition.

We assume that the BoJ's ETF purchasing policy rule takes the following form, in

[1] We assume that the potential outcome depends only on past and contemporaneous assignments. Rambachan and Shephard (2021) called this assumption Non-anticipating potential outcomes.

which the amount of ETF purchases at time t, ETF_t, is given by

$$ETF_t = ETF_{-,t}(\Delta p_t)\mathbb{I}(\Delta p_t < c_t) + ETF_{+,t}(\Delta p_t)\mathbb{I}(\Delta p_t \geq c_t), \tag{A.1}$$

where Δp_t is the log change in the TOPIX value in the morning, c_t is the cut-off, and $ETF_{-,t}$ and $ETF_{+,t}$ are random functions of Δp_t which represent different policy rules depending on whether Δp_t is above or below the cutoff at time t. The following assumptions guarantee that our estimands identify the dynamic local average treatment effect.

Assumption 1. *(i)* $Y_{t+l,h}(w)$ *is bounded and continuously differentiable in* $w \in [0, \bar{w}]$ *with probability one, and (ii)* $ETF_{-,t}(\Delta p)$ *and* $ETF_{+,t}(\Delta p)$ *are bounded and continuous at* c_t *with probability one.*

Assumption 2 (Monotonicity). $ETF_{-,t}(c_t) \geq ETF_{+,t}(c_t)$ *with probability one.*

Assumption 3 (Relevance). $\int \Pr(ETF_{-,t}(c_t) \geq w \geq ETF_{+,t}(c_t)|\Delta p_t = c_t)dw > 0$.

Assumption 4 (Local Independence). *For each* $t + l$ *and* h*, there exists a neighborhood* $N_{t+l,h}$ *of* c_t *such that* $\Delta p_t \perp (\{Y_{t+l,h}(w)\}_w, ETF_{-,t}(c_t), ETF_{+,t}(c_t))|\Delta p_t \in N_{t+l,h}$.

Theorem 1. *If Assumptions 1-4 hold, then*

$$\begin{aligned}
&\frac{\lim_{\Delta p \uparrow c_t} \mathbb{E}[Y_{t+l,h}|\Delta p_t = \Delta p] - \lim_{\Delta p \downarrow c_t} \mathbb{E}[Y_{t+l,h}|\Delta p_t = \Delta p]}{\lim_{\Delta p \uparrow c_t} \mathbb{E}[ETF_t|\Delta p_t = \Delta p] - \lim_{\Delta p \downarrow c_t} \mathbb{E}[ETF_t|\Delta p_t = \Delta p]} \\
&= \int \mathbb{E}[\frac{\partial Y_{t+l,h}(w)}{\partial w}|\Delta p_t = c_t, ETF_{-,t}(c_t) \geq w \geq ETF_{+,t}(c_t)]\bar{\omega}dw,
\end{aligned}$$

where $\bar{\omega} = \Pr(ETF_{-,t}(c_t) \geq w \geq ETF_{+,t}(c_t)|\Delta p_t = c_t) / \int \Pr(ETF_{-,t}(c_t) \geq w \geq ETF_{+,t}(c_t)|\Delta p_t = c_t)dw$.

Proof. First, observe that

$$\begin{aligned}
\lim_{\Delta p \uparrow c_t} \mathbb{E}[Y_{t+l,h}|\Delta p_t = \Delta p] &= \lim_{\Delta p \uparrow c_t} \mathbb{E}[Y_{t+l,h}(ETF_{-,t}(\Delta p))|\Delta p_t = \Delta p] \\
&= \mathbb{E}[Y_{t+l,h}(ETF_{-,t}(c_t))|\Delta p_t = c_t],
\end{aligned}$$

and

$$\lim_{\Delta p \downarrow c_t} \mathbb{E}[Y_{t+l,h} | \Delta p_t = \Delta p] = \lim_{\Delta p \downarrow c_t} \mathbb{E}[Y_{t+l,h}(ETF_{+,t}(\Delta p)) | \Delta p_t = \Delta p]$$
$$= \mathbb{E}[Y_{t+l,h}(ETF_{+,t}(c_t)) | \Delta p_t = c_t],$$

follow from Assumptions 1 and 4. Therefore,

$$\begin{aligned}
&\lim_{\Delta p \uparrow c_t} \mathbb{E}[Y_{t+l,h} | \Delta p_t = \Delta p] - \lim_{\Delta p \downarrow c_t} \mathbb{E}[Y_{t+l,h} | \Delta p_t = \Delta p] \\
&= \mathbb{E}[Y_{t+l,h}(ETF_{-,t}(c_t)) - Y_{t+l,h}(ETF_{+,t}(c_t)) | \Delta p_t = c_t] \\
&= \mathbb{E}[\int \frac{\partial Y_{t+l,h}(w)}{\partial w} \mathbb{I}\{ETF_{-,t}(c_t) \geq w \geq ETF_{+,t}(c_t)\} dw | \Delta p_t = c_t] \\
&= \int \mathbb{E}\left[\frac{\partial Y_{t+l,h}(w)}{\partial w} | \Delta p_t = c_t, ETF_{-,t}(c_t) \geq w \geq ETF_{+,t}(c_t)\right] \\
&\quad \times \Pr(ETF_{-,t}(c_t) \geq w \geq ETF_{+,t}(c_t) | \Delta p_t = c_t) dw,
\end{aligned}$$

where the second equality follows from Assumptions 1 and 2, and the third equality follows from Assumption 1. Similarly,

$$\lim_{\Delta p \uparrow c_t} \mathbb{E}[ETF_t | \Delta p_t = \Delta p] - \lim_{\Delta p \downarrow c_t} \mathbb{E}[ETF_t | \Delta p_t = \Delta p]$$
$$= \int \Pr(ETF_{-,t}(c_t) \geq w \geq ETF_{+,t}(c_t) | \Delta p_t = c_t) dw,$$

and Assumption 3 guarantees that the denominator is positive. Combining these, we have the stated result. □

Since $Y_{t+l,h}(w) \equiv Y_{t+l,h}(ETF_{1:t-1}, w, ETF_{t+1:t+l})$, the local independence assumption requires that falling below the cutoff on day t is not correlated with future or past ETF purchases. We test this in Appendix A.4.

A.2 Details on Cutoff Estimation

We first split the sample at six BoJ ETF policy announcements: April 4, 2013, October 31, 2014, December 18, 2015, July 29, 2016, July 31, 2018, and March 16, 2020, as summarized in Table B.1. The announcements changed purchase amounts or implementation details. In July 2018, purchase flexibility and allocation changed while the overall annual pace remained unchanged. We then divide each sample according to whether the TOPIX close-to-close return is negative on each of the preceding two trading days.

In each sample split, we proceed as follows. We take grid points for the cutoff candidates from -1% to 0% at 0.05% intervals, $\mathbb{C} = \{-1.0\%, -0.95\%, \ldots, -0.05\%, 0.0\%\}$. For each $c \in \mathbb{C}$, we estimate the following linear probability model separately on both sides of the candidate cutoff, c:

$$\Pr_{-,t}(ETF_t > 0 | \Delta p_t) = \begin{cases} \alpha_- + \beta_- \Delta p_t & \text{for } \Delta p_t \in [c-k, c] \\ \alpha_+ + \beta_+ \Delta p_t & \text{for } \Delta p_t \in [c, c+k] \end{cases}, \tag{A.2}$$

where we take the bandwidth to be 1% around the cutoff, $k = 1\%$. Given the estimates, we can compute the jump around the cutoff as follows:

$$J_t(c) \equiv \lim_{\Delta p \downarrow c} \widehat{\Pr_t}(ETF_t > 0 | \Delta p) - \lim_{\Delta p \uparrow c} \widehat{\Pr_t}(ETF_t > 0 | \Delta p),$$

where $\widehat{\Pr_t}$ denotes the fitted value of equation (A.2). We select the cutoff that maximizes the square of the jump:

$$c_t^* \in \arg\max_{c \in \mathbb{C}} J_t^2(c).$$

We treat candidate cutoffs whose squared jumps differ from the maximum by less than 0.01 as tied. For subsamples without two consecutive declines, we choose the candidate that also belongs to the near-maximizing candidate sets in the greatest number of adjacent time periods.

Table B.2 shows the estimated cutoff, and Table B.3 shows the discontinuity in the probability of the Bank of Japan's intervention around the estimated cutoff. As argued in the main text, the estimated cutoffs align well with what is commonly reported in the media. The discontinuity around the cutoff always exceeds 50 percentage points in magnitude, often exceeds 80 percentage points, and is highly statistically significant.

Figure B.1 graphically displays the discontinuity in the probability of intervention for each period. While the magnitude of the discontinuity is more apparent at the beginning and the end of the sample period, the sharp discontinuity shows up in all subsamples.

A.3 Manipulation Test

A typical concern in regression discontinuity-based identification strategies is manipulation (McCrary, 2008). We formally test for the presence of manipulation by examining the continuity of the density function of TOPIX changes in the morning. We estimate the density function using the local polynomial density estimator by Cattaneo et al. (2020) and test for the presence of discontinuity around our estimated cutoff.

Figure 2C shows the estimated density and histogram, and the benchmark column of Table B.4 reports the estimates and test statistics for discontinuity. While there is a small mass on the right side of the cutoff, the p-value for the discontinuity test is 0.373. Therefore, there is no statistical evidence of manipulation.

We further examine the robustness of this result to alternative bandwidth choices. Table B.4 reports the density discontinuity test using bandwidths ranging from 0.5 to 1.5 times the benchmark bandwidth. We continue to find no statistically significant discontinuity across these alternative bandwidth choices. Moreover, when we use narrower bandwidths to focus more closely on observations around the cutoff, the estimated discontinuity becomes smaller rather than larger compared to the benchmark. This is particularly informative for the visual concern in Figure 2C, since narrower bandwidths make the estimation more local to the cutoff, where the apparent discontinuity would arise. In summary, there appears to be no evidence of manipulation around the cutoff.

A.4 (Dis)continuity of ETF Purchases across Days

In this section, we argue that the effects we are identifying are the effects of a one-time shock to ETF purchases. As discussed in Appendix A.1, $y_{t+l,h}$ is clearly affected by the BoJ's ETF purchases up to l days later. Therefore, if falling below the cutoff today is correlated with future and past purchases, our empirical estimates cannot be interpreted as the causal effect of one-time BoJ ETF purchases (Rambachan and Shephard, 2021). In order to address this concern, we estimate the discontinuity in the amount of ETF purchases around the cutoff across days. Formally, we estimate the following term:

$$\lim_{\Delta p \uparrow c_t} \mathbb{E}[ETF_{t+l} | \Delta p_t = \Delta p] - \lim_{\Delta p \downarrow c_t} \mathbb{E}[ETF_{t+l} | \Delta p_t = \Delta p]. \tag{A.3}$$

Figure B.2 shows the estimates of discontinuity in the amount of ETF purchases at date $t+l$ around $\Delta p_t = c_t$. Reassuringly, we find a significant discontinuity only at $l = 0$. This finding supports interpreting our estimates as the effects of a one-time BoJ ETF purchase, rather than effects driven by associated future or past purchases.

B Additional Tables and Figures

Table B.1: ETF Policy Announcements

Date	Announcement
April 4, 2013	Annual increase in ETF holdings set at 1 trillion yen
October 31, 2014	Annual increase in ETF holdings raised to about 3 trillion yen
December 18, 2015	Additional annual ETF purchase program of 300 billion yen announced, starting in April 2016
July 29, 2016	Annual increase in ETF holdings raised to about 6 trillion yen
July 31, 2018	More flexible purchases and increased TOPIX allocation
March 16, 2020	Temporary annual upper limit set at about 12 trillion yen

Notes: Table B.1 shows the six major announcements by the BoJ regarding the ETF purchase amounts used to define the sample split. Source: Bank of Japan website.

Table B.2: Estimated Cutoff

No Consecutive Drops		Consecutive Drops	
Period	Cutoff	Period	Cutoff
2010/12/15 - 2013/04/03	-1%	2010/12/15 - 2013/04/03	-1%
2013/04/04 - 2014/10/30	-0.35%	2013/04/04 - 2014/10/30	0%
2014/10/31 - 2015/12/17	-0.15%	2014/10/31 - 2015/12/17	0%
2015/12/18 - 2016/07/28	-0.4%	2015/12/18 - 2016/07/28	0%
2016/07/29 - 2018/07/30	-0.3%	2016/07/29 - 2018/07/30	0%
2018/07/31 - 2020/03/15	-0.5%	2018/07/31 - 2020/03/15	-0.25%
2020/03/16 - 2020/12/31	-0.5%	2020/03/16 - 2020/12/31	-0.25%

Notes: Table B.2 shows the estimated cutoff for each of the subsamples.

Table B.3: Discontinuity in Probability of Intervention around the Estimated Cutoff

	No Consecutive Drops			Consecutive Drops		
	Discontinuity estimates	Sample size Left	Right	Discontinuity estimates	Sample size Left	Right
2010/12/15 - 2013/04/03	-1.011	43	158	-1.000	15	47
	(0.012)			(0.000)		
2013/04/04 - 2014/10/30	-0.576	60	147	-0.931	25	30
	(0.100)			(0.072)		
2014/10/31 - 2015/12/17	-0.683	72	98	-1.122	11	17
	(0.099)			(0.117)		
2015/12/18 - 2016/07/28	-0.814	21	36	-1.000	8	11
	(0.118)			(0.000)		
2016/07/29 - 2018/07/30	-0.608	78	244	-0.945	40	33
	(0.069)			(0.053)		
2018/07/31 - 2020/03/15	-0.978	49	163	-0.744	30	34
	(0.022)			(0.130)		
2020/03/16 - 2020/12/31	-0.930	29	71	-0.985	13	14
	(0.070)			(0.022)		

Notes: Table B.3 shows the discontinuity in the probability of the BoJ's intervention around the estimated cutoff. We report right-minus-left discontinuities: the fitted probability just above the cutoff minus that just below it. We estimate the discontinuity using local linear regression with a bandwidth of 1% around the cutoff and a uniform kernel. The standard errors are reported in parentheses.

Table B.4: Density Discontinuity Test

	Bandwidth				
	Benchmark	$\times 0.50$	$\times 0.75$	$\times 1.25$	$\times 1.50$
Density estimates					
Left	0.423	0.412	0.439	0.444	0.455
Right	0.506	0.472	0.474	0.520	0.526
Discontinuity test					
Difference	0.083	0.060	0.035	0.077	0.071
	(0.093)	(0.129)	(0.107)	(0.082)	(0.074)
95% CI	$[-0.099, 0.265]$	$[-0.194, 0.314]$	$[-0.174, 0.244]$	$[-0.083, 0.236]$	$[-0.074, 0.216]$
p-value	0.373	0.642	0.743	0.348	0.338
Bandwidth					
Left	0.512	0.256	0.384	0.639	0.767
Right	0.620	0.310	0.465	0.775	0.930
Effective sample size					
Left	358	225	295	404	443
Right	719	378	545	892	1044

Notes: The table reports density-discontinuity estimates at the estimated cutoff, using the local polynomial procedure of Cattaneo et al. (2020), with order $p = 2$, bias-correction order $q = 3$, a triangular kernel, unrestricted fits, and jackknife standard errors. The benchmark uses the optimal bandwidth. Other columns multiply both benchmark bandwidths by the indicated factor. "Difference" is the right density minus the left density, with standard error is in parentheses. The confidence intervals are pointwise 95% intervals for this difference. The full sample contains 667 observations to the left and 1,790 to the right of the cutoff.

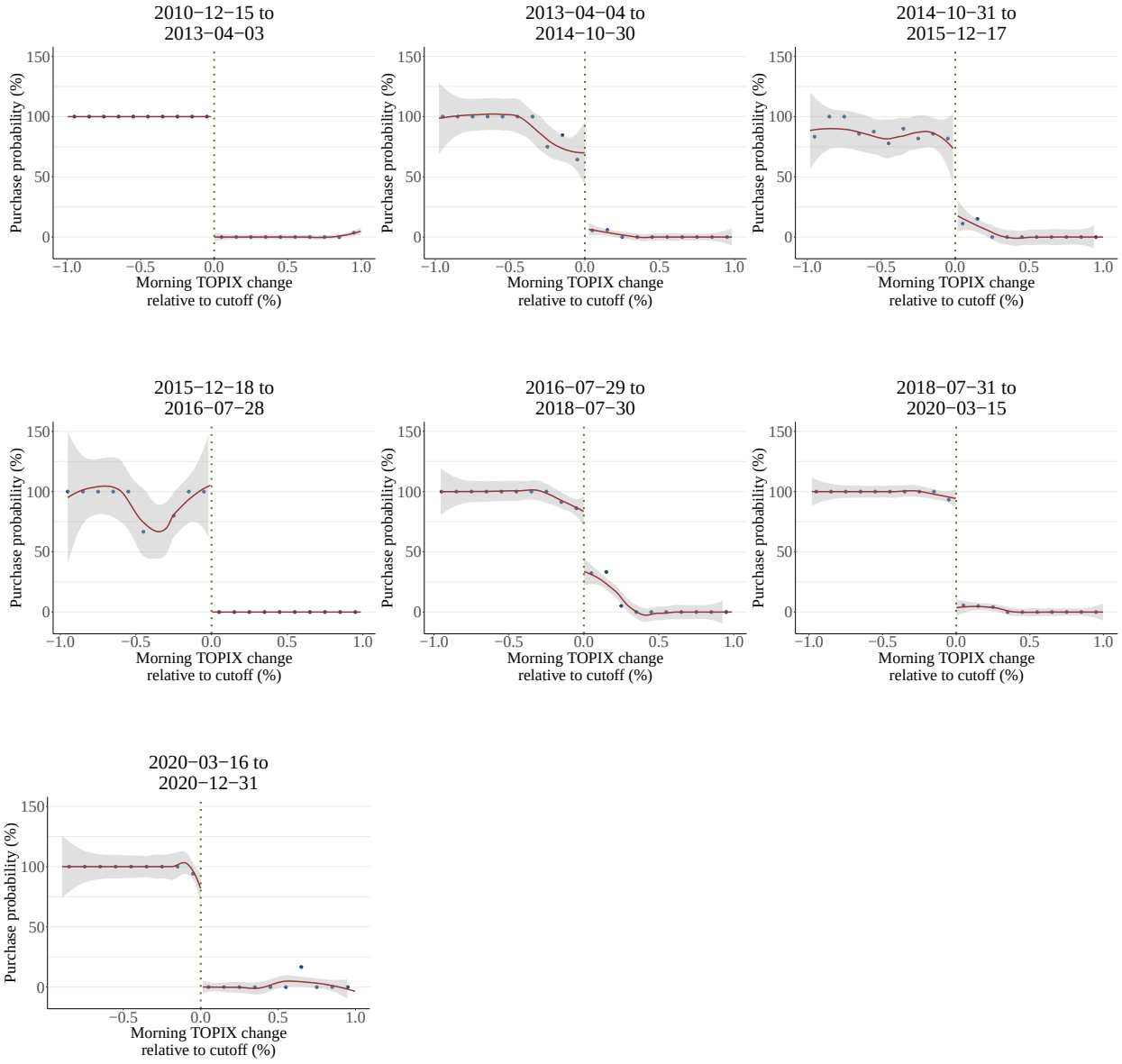


Figure B.1: Discontinuity in the Probability of the BoJ's Intervention for each Period

Notes: Figure B.1 shows the discontinuity in the probability of the BoJ's intervention around the estimated cutoff for each period. Each blue dot shows the percentage of days with an intervention within a bin of width 0.1 percentage point. The red lines show separate LOESS fits on each side of the cutoff, and the shaded gray areas indicate 95% confidence intervals.

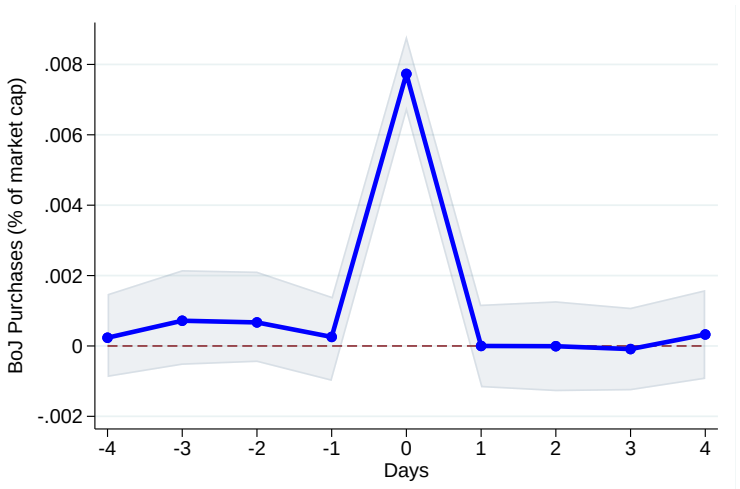


Figure B.2: (Dis)continuity of ETF Purchases across Days

Notes: Figure B.2 shows the estimates of (A.3) across days. The shaded areas represent 90% confidence intervals, which account for heteroskedasticity and autocorrelation.

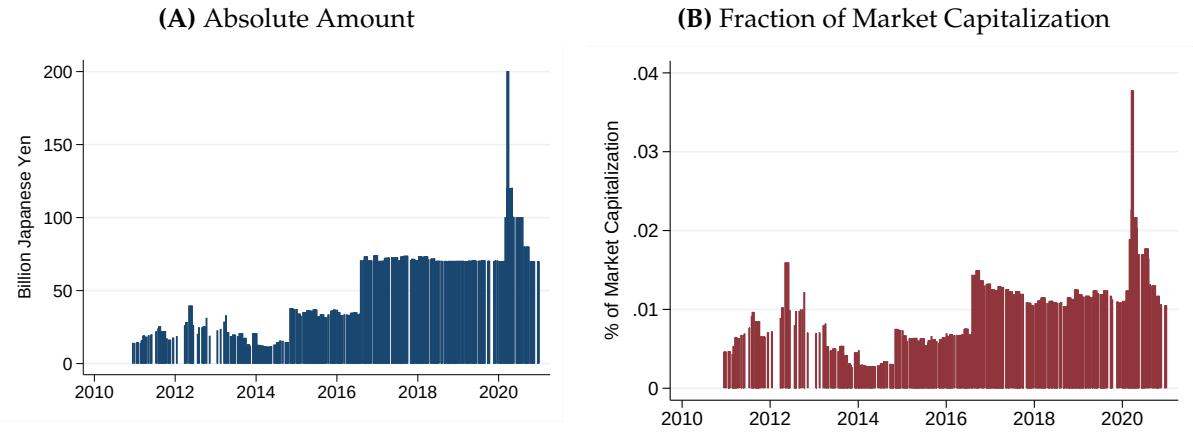


Figure B.3: The Amount of BoJ Purchases

Notes: Figure B.3 plots the amount of stock purchases by the BoJ in each intervention. Figure B.3A shows the absolute amount of purchases in billion Japanese yen (approximately 10 million US dollars). Figure B.3B expresses it as a fraction of market capitalization.

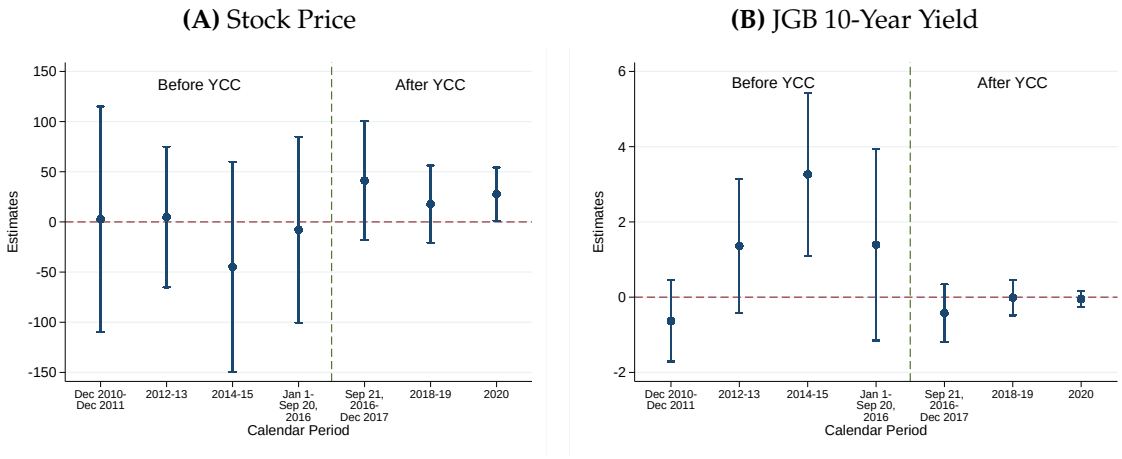


Figure B.4: Heterogeneity Around the Introduction of Yield Curve Control: Disaggregated Periods

Notes: Panels A and B show, respectively, the responses of stock prices and 10-year JGB yields, divided into seven calendar periods relative to the introduction of YCC on September 21, 2016. The four periods before YCC are December 2010–December 2011, January 2012–December 2013, January 2014–December 2015, and January 1–September 20, 2016. The three periods after YCC are September 21, 2016–December 2017, January 2018–December 2019, and January 2020–December 2020. The estimates measure the changes from 11 AM on the intervention day to 9 AM on the next trading day. The 90% confidence intervals account for heteroskedasticity and autocorrelation.

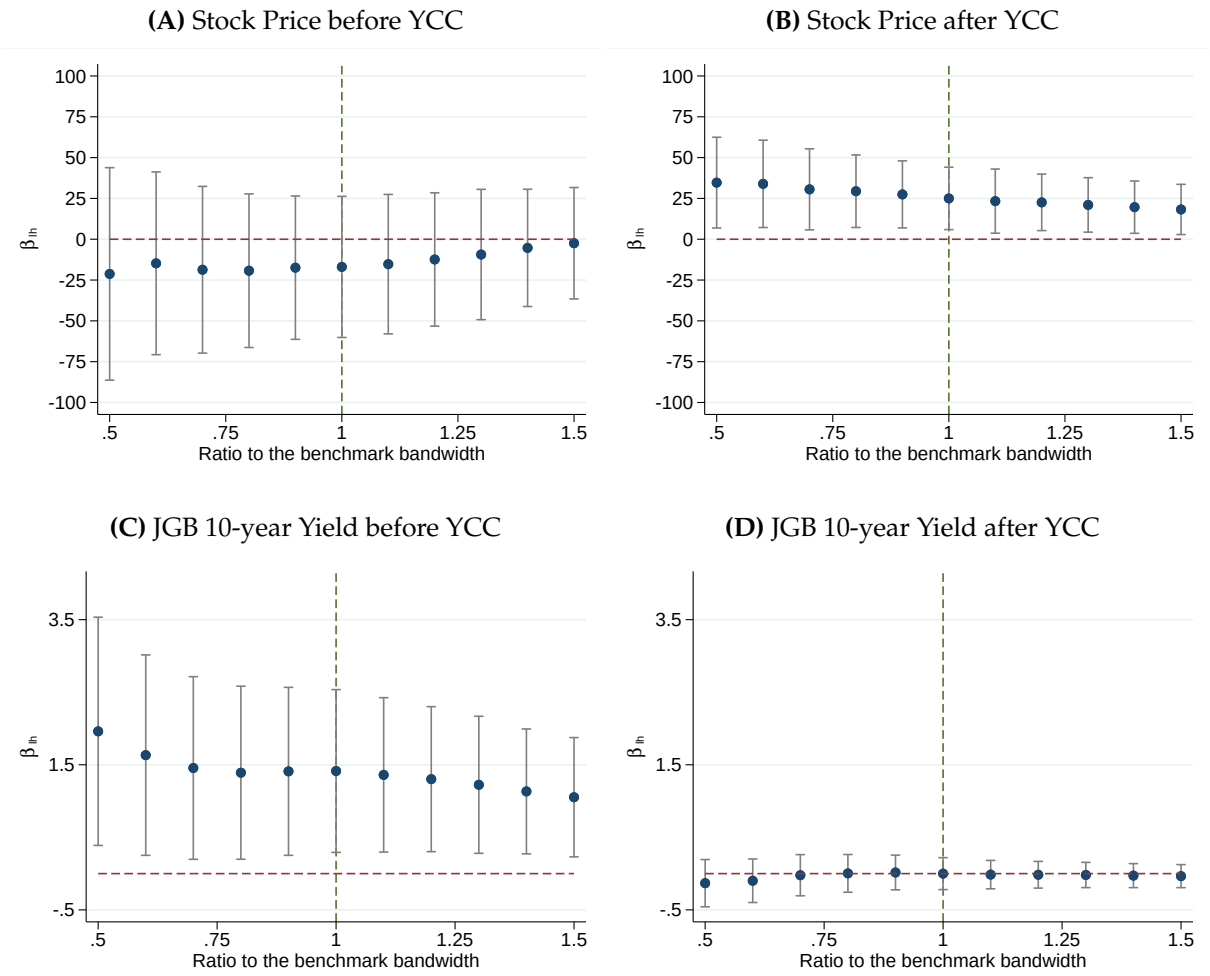


Figure B.5: Robustness to Bandwidth Selection

Notes: Figure B.5 shows the robustness of our estimates with respect to the size of the bandwidth. Each dot represents the point estimate of the response from 11 AM on the intervention day to 9 AM on the next day. The vertical line represents the 90% confidence interval, which accounts for heteroskedasticity and autocorrelation. The dashed green line is the optimal bandwidth proposed by Calonico, Cattaneo, and Titiunik (2014), which is our benchmark. Figures B.5A and B.5B show the response of stock prices before and after YCC, respectively. Figures B.5C and B.5D show the response of the 10-year JGB yield before and after YCC, respectively.

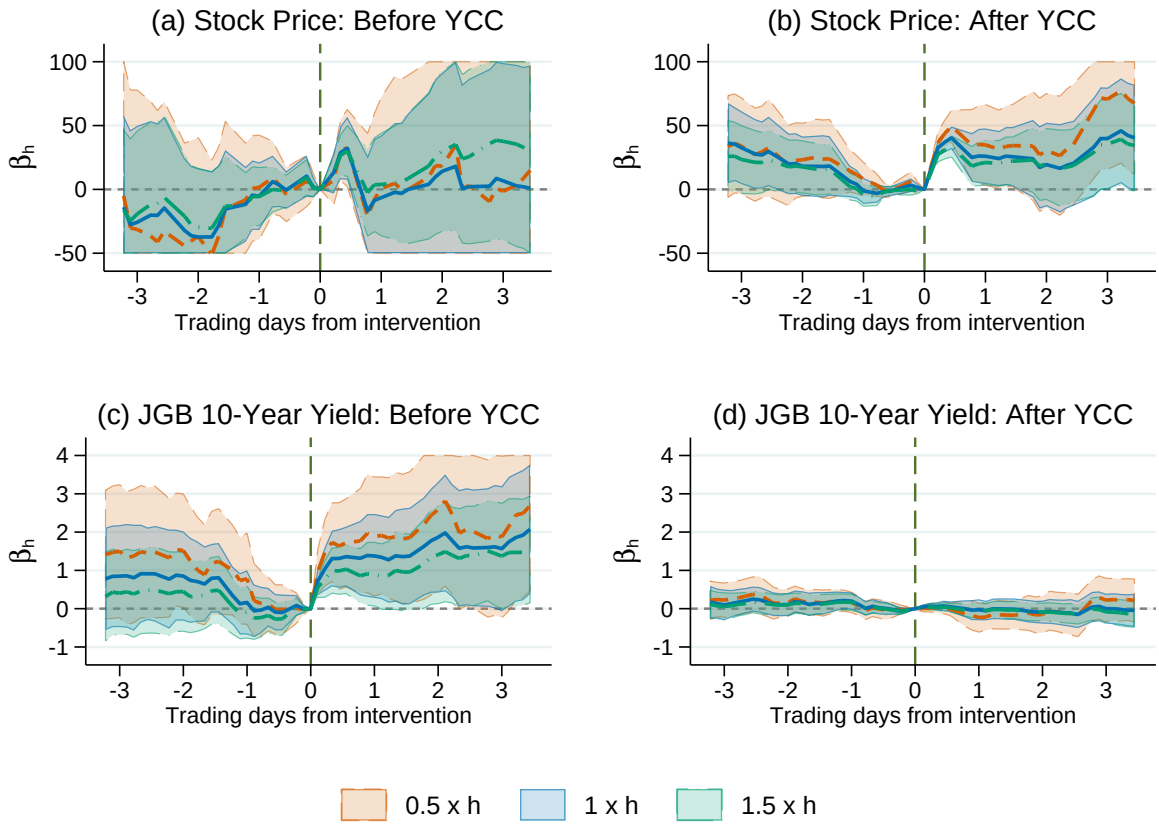


Figure B.6: Pre-Trends

Notes: Panels (a) and (b) show the impulse responses of stock prices before and after YCC, respectively, by plotting the coefficient $\beta_{l,h}$ in equation (1). Panels (c) and (d) show the corresponding responses of the 10-year JGB yield. The stock price coefficient measures the log changes in stock prices in response to stock purchases of 1% of market capitalization, and the yield coefficient measures the percentage point changes in the yield in response to stock purchases of 1% of market capitalization. Orange, blue, and teal lines use bandwidths of $0.5h$, h, and $1.5h$, respectively, where h is the optimal bandwidth. The shaded areas represent pointwise 90% confidence intervals that account for heteroskedasticity and autocorrelation.

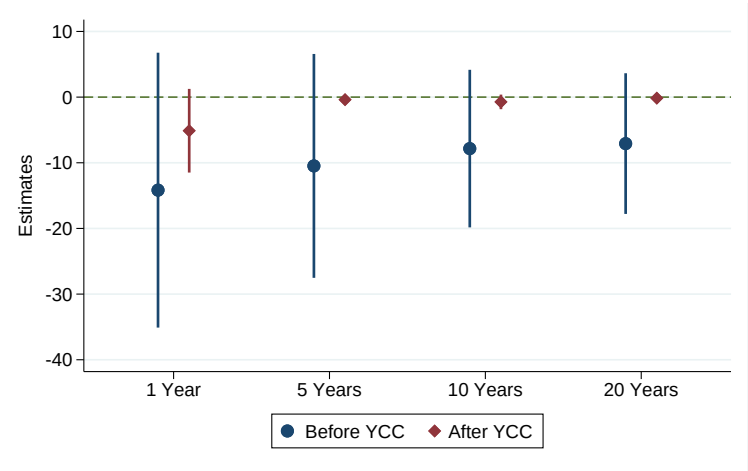


Figure B.7: Response of Inflation Swaps

Notes: Figure B.7 shows the response of the inflation swap rate across different maturities from 11 AM on the day of the intervention to 9 AM on the next day. The circles represent the point estimates before yield curve control, and the diamonds represent the point estimates after yield curve control. The coefficient measures the percentage point changes in the inflation swap rate in response to the purchase of 1% of market capitalization. The vertical lines represent the 90% confidence intervals, which account for heteroskedasticity and autocorrelation. We obtained the tick-by-tick inflation swap rate from Bloomberg.

C Placebo Tests

One might worry that our results are not driven by the discontinuous changes in the BoJ's policy intervention, but rather by some other factors such as investors' sentiments that sharply respond to the stock price changes in the morning session. Some investors might employ simple trading strategies that respond *discontinuously* to stock price changes during the morning session, independent of the BoJ's interventions (e.g., because of behavioral heuristics among investors). For instance, if stock prices drop by more than 1% in the morning, investors may heuristically anticipate a market rebound and adjust their portfolios accordingly, regardless of any potential BoJ actions. This kind of behavior introduces a separate, discontinuous reaction that coincides with the BoJ's intervention threshold, which could potentially confound our RDD estimates.

To address this concern, we conduct placebo tests by testing for the presence of discontinuity in outcome variables around an arbitrary cutoff for which we do not expect to find any discontinuity. Specifically, for each placebo cutoff $c^{placebo} \in \{-1\%, -0.95\%, \ldots, -0.05\%, 0\%\}$, we test whether there is a discontinuity in our outcome variables when the stock prices fall below the placebo cutoff $c^{placebo}$. Namely, we estimate

$$\gamma_{l,h} \equiv \lim_{\Delta p \uparrow c^{placebo}} \mathbb{E}[\Delta y_{t+l,h} | \Delta p_t = \Delta p] - \lim_{\Delta p \downarrow c^{placebo}} \mathbb{E}[\Delta y_{t+l,h} | \Delta p_t = \Delta p], \tag{C.1}$$

where Δp_t is the percentage change in TOPIX in the morning session, and we focus on within-day changes (from 11 AM to 3 PM) in the outcome variables. Importantly, when estimating $\gamma_{l,h}$, we exclude periods for which the actual cutoff is identical to the placebo cutoff $c^{placebo}$ under consideration. We are interested in the estimates of $\gamma_{l,h}$, and we expect $\gamma_{l,h}$ to be indistinguishable from zero for any value of $c^{placebo}$.

Figures C.8A and C.8B show the estimated values of $\gamma_{l,h}$ across the placebo cutoffs, as well as the baseline estimates (red lines) obtained by using the true cutoff c_t instead of the placebo cutoff $c^{placebo}$ in Equation (C.1): i.e., $\lim_{\Delta p \uparrow c_t} \mathbb{E}[\Delta y_{t+l,h} | \Delta p_t = \Delta p] - \lim_{\Delta p \downarrow c_t} \mathbb{E}[\Delta y_{t+l,h} | \Delta p_t = \Delta p]$. Reassuringly, while the baseline estimates are statistically significant, we find that the estimates of $\gamma_{l,h}$ are indistinguishable from zero in almost all cases. Even in cases where the estimates are significant, they consistently exhibit the opposite sign compared to the baseline estimates. Moreover, in every case, the estimates deviate substantially from the baseline estimates that use the actual cutoff. These results strongly suggest that our results are indeed driven by the BoJ's policy intervention rather than by unrelated investor behavior.

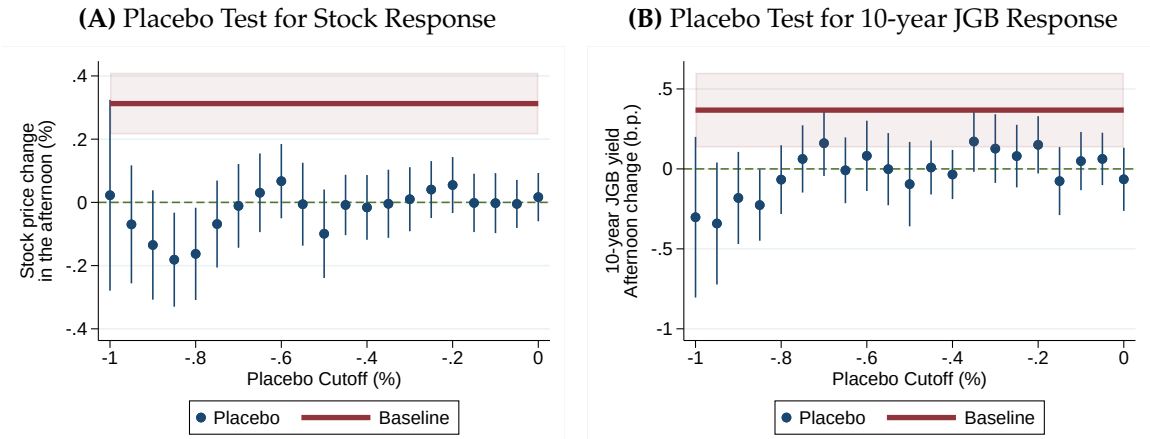


Figure C.8: Placebo Tests

Notes: Figure C.8A plots the estimates of $\gamma_{l,h}$ in equation (C.1) as the blue dots for each placebo cutoff, where the outcome variable is the stock price. The estimates are the response within the day of intervention (changes from 11 AM to 3 PM). We exclude the periods where the placebo cutoff coincides with the actual cutoff. The red line indicates our estimates using the actual cutoff. Figure C.8B is analogous to Figure C.8A, where the outcome variable is now the 10-year JGB yield. The line and the shaded area represent 90% confidence intervals.

D JGB Purchases at the ETF Purchase Cutoff

A potential concern is that the response of the long-term interest rate that we document is driven by the BoJ's intervention in the Japanese government bond (JGB) market that occurs at the same time as the stock purchases. First, as we mention in Section 4.4 in the main text, the BoJ sold JGBs only twice, on March 24, 2017, and March 23, 2020, both of which were after yield curve control. Therefore, the rise in interest rates before YCC cannot be driven by the BoJ selling JGBs.

Second, we address the concern that the interest rate response might reflect the BoJ systematically changing the size of JGB purchases at the same time as the stock purchases. This is a legitimate concern, since the BoJ regularly purchased JGBs throughout our sample period. We examine this concern using daily outright JGB purchase amounts constructed from the BoJ's market operation reports. We record successful bid amounts by offer date, in billion yen, and assign zero to trading days without purchases.

Figure D.9 shows aggregate purchases separately before and after YCC. Figures D.10 and D.11 examine four groups based on remaining maturity. The aggregate series uses our full sample; the maturity breakdown begins in June 2013, when comparable maturity categories become available.

From these figures, we can first visually confirm that there is no discontinuity in JGB purchases at the ETF purchase cutoff. In fact, the estimated aggregate discontinuity, measured as purchases below minus above the cutoff, is -0.4 billion yen before YCC (standard error 80.3) and -20.9 billion yen after YCC (standard error 54.5). Neither aggregate estimate nor any of the eight estimates by maturity is statistically significant at the conventional level. These baseline tests therefore do not reveal a discontinuous change in JGB purchase amounts at the ETF purchase cutoff.

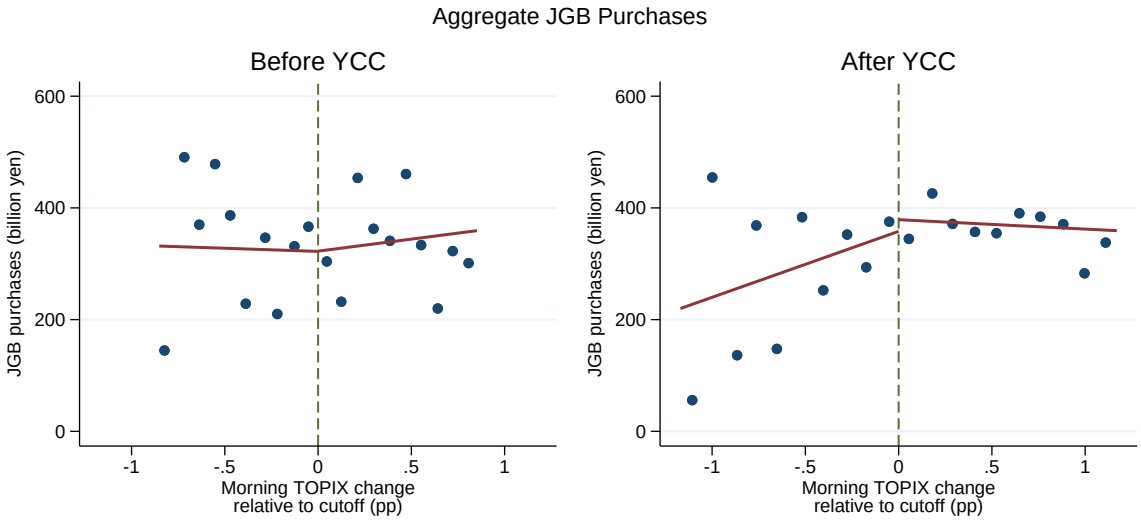


Figure D.9: JGB Purchases around the ETF Purchase Cutoff

Notes: The panels show daily aggregate outright JGB purchases before and after YCC. The horizontal axis is the morning TOPIX change relative to the estimated ETF purchase cutoff. Each dot plots mean purchases against the mean morning TOPIX change relative to the cutoff within a bin, using ten equal-width bins on each side of the cutoff within the selected bandwidth. Lines are separate local-linear fits with a triangular kernel and an outcome-specific bandwidth. Standard errors reported in the text account for heteroskedasticity and autocorrelation.

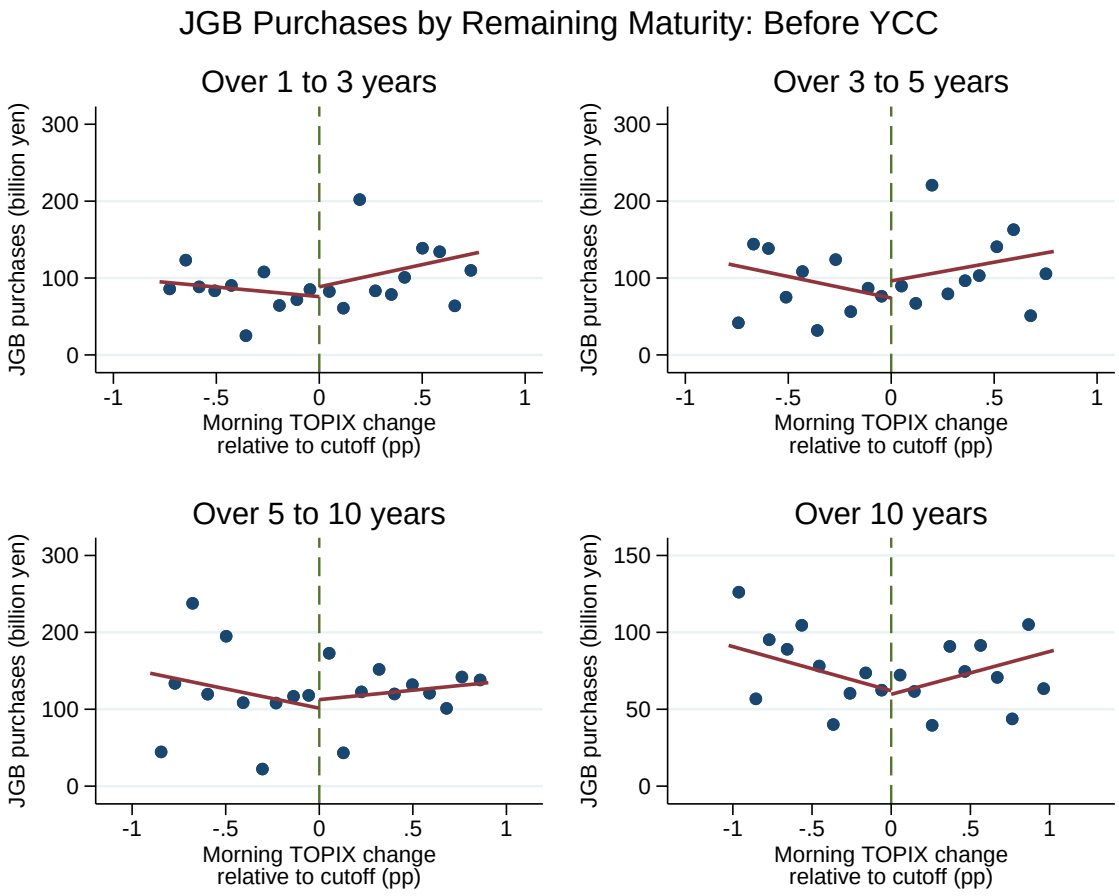


Figure D.10: JGB Purchases by Remaining Maturity: Before YCC

Notes: The panels show daily outright JGB purchases before YCC for four remaining-maturity groups: over 1 to 3 years, over 3 to 5 years, over 5 to 10 years, and over 10 years. Dots and fitted lines are constructed as in Figure D.9, using a separate bandwidth for each group. Each group uses the same axis limits in the before- and after-YCC figures.

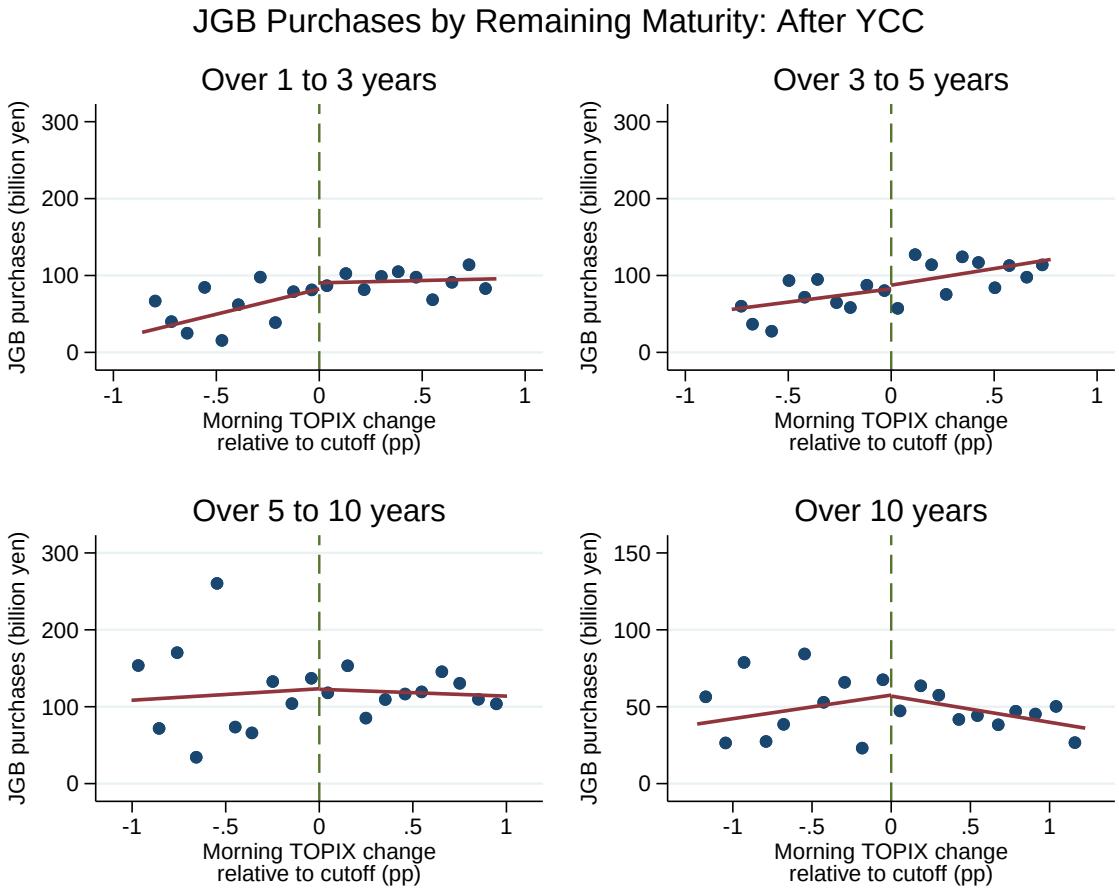


Figure D.11: JGB Purchases by Remaining Maturity: After YCC

Notes: The panels show daily outright JGB purchases after YCC for the same four remaining-maturity groups as Figure D.10. Dots and fitted lines are constructed as in Figure D.9, using a separate bandwidth for each group. Axis limits match those in Figure D.10.

E Bond Yield Responses Across Different Maturities

We show that the effect on the interest rate is not specific to the 10-year JGB yield, but rather is widespread across the entire yield curve.

Figure E.12 shows the point estimates of the effect on the JGB yield across maturities of 1, 5, 10, 20, and 30 years. Before yield curve control, yields rose at all maturities, but the effect was larger at longer maturities. Our preferred interpretation is that the zero lower bound on the policy rate was binding during this period, and therefore the shorter-maturity bonds had less room to respond relative to longer-maturity bonds. After yield curve control, all interest rates entirely stopped responding. Although yield curve control was intended to specifically control the 10-year yield, it can prevent the other maturities from responding because they are interconnected through arbitrage. For example, this is a natural prediction of the preferred habitat model of the term structure by Vayanos and Vila (2021).

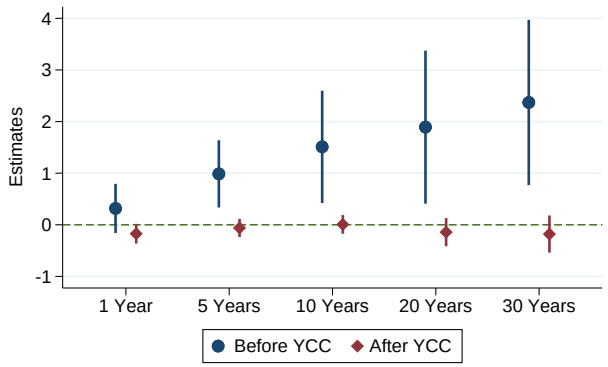


Figure E.12: Response of JGB Yields Across Maturities

Notes: The figure shows the response of the JGB yield across different maturities from 11 AM on the day of the intervention to 9 AM on the next day. The circles represent the point estimates before yield curve control, and the diamonds represent the point estimates after yield curve control. The coefficient measures the percentage point changes in the JGB yield in response to the purchase of 1% of market capitalization. All confidence intervals account for heteroskedasticity and autocorrelation.

F Intraday Responses of TOPIX Futures

In the baseline analysis, we focused on the spot TOPIX index as a measure of the stock price. Since the stocks are not traded during the lunch break, we are not able to see the price adjustment during the lunch break, during which the BoJ is considered to have intervened.

In this appendix, we examine TOPIX futures responses around the stock market's morning close. When futures trading remains open during the stock market's lunch break, futures prices allow us to observe price adjustment before afternoon stock trading resumes. We obtain the TOPIX futures prices from the Japan Exchange Group Data Cloud.

Before November 21, 2011, both the spot and the futures markets had a lunch break from 11 AM to 12:30 PM. On that date, the stock market extended its morning session to 11:30 AM, while retaining the 12:30 PM afternoon opening. TOPIX futures held a morning closing auction at 11:35 AM and resumed regular trading at 11:45 AM, allowing us to observe futures prices during much of the stock market's lunch break. On March 24, 2014, the Tokyo Stock Exchange's derivatives market was integrated into Osaka Exchange's J-GATE platform, and since then, futures have been traded continuously throughout the stock market's lunch break.

Figures F.13 and F.14 plot the impulse response of TOPIX futures at five-minute intervals around the stock market's morning close, by plotting the coefficient $\beta_{l,h}$ in equation (1). Figure F.13 pools the available pre-YCC subperiods by estimating relative to the closing time of the morning session. We find that stock price futures respond during the lunch break both before and after YCC, although the response is larger and more precisely estimated after YCC. Figure F.14 divides the pre-YCC periods into three subperiods, with different trading hours, and we find similar patterns.

Overall, the timing of the futures-price response is consistent with investors anticipating the BoJ's intervention.

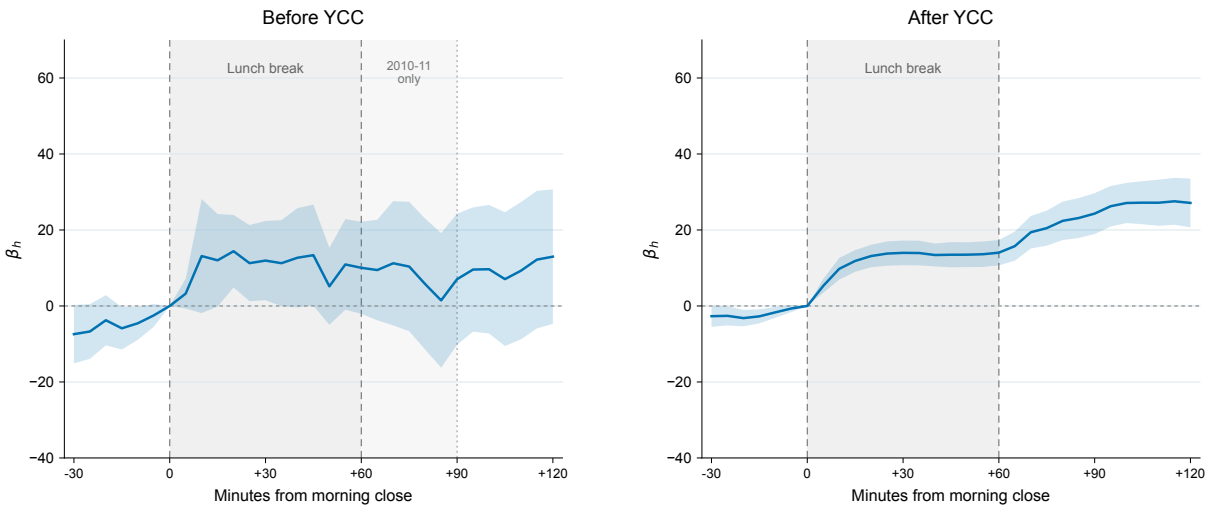


Figure F.13: TOPIX Futures Responses Before and After YCC

Notes: The panels show the impulse response of TOPIX futures at five-minute intervals around the stock market's morning close by plotting the coefficient $\beta_{l,h}$ in equation (1). Responses are relative to the five-minute window ending at the stock market's morning close: 11 AM before November 21, 2011, and 11:30 AM thereafter. The horizontal axis measures minutes from the closing time of the morning session. Gray shading denotes the stock market's lunch break, and the light gray shading denotes the lunch break only for the period before November 21, 2011. Blue shaded areas show 90% confidence intervals accounting for heteroskedasticity and autocorrelation.

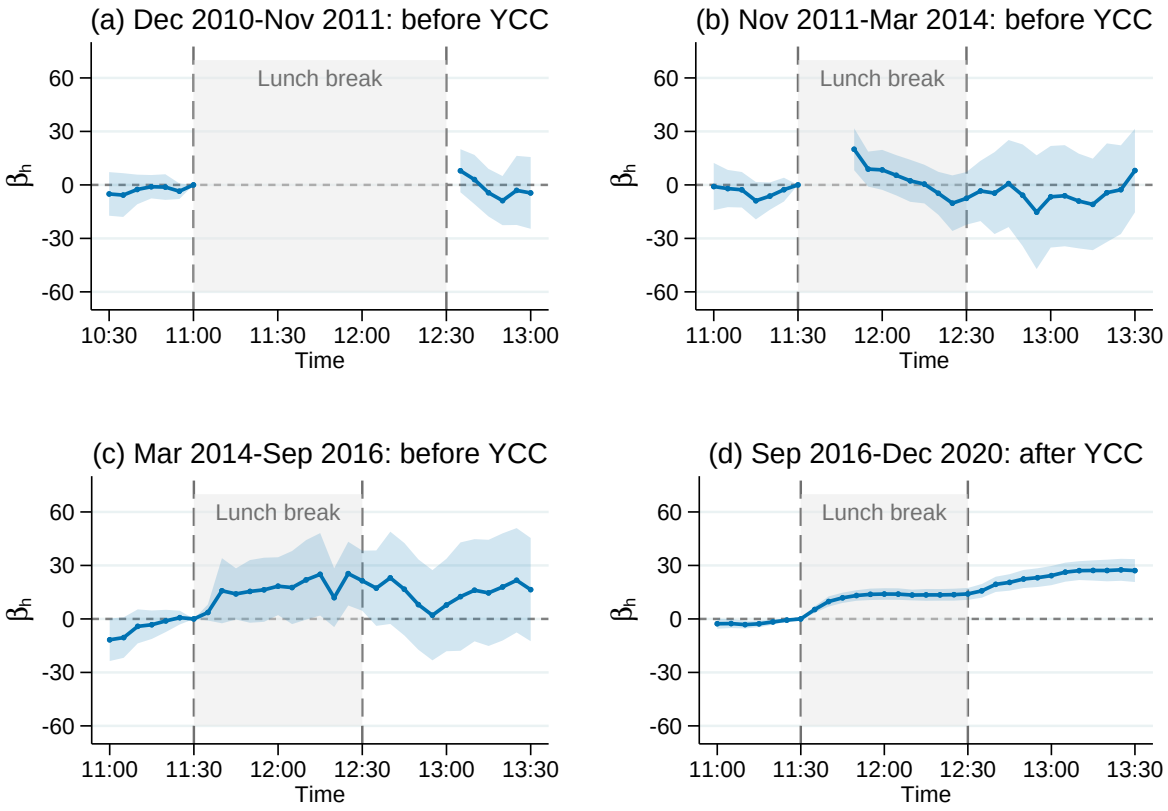


Figure F.14: TOPIX Futures Responses by Subperiod

Notes: The figure is analogous to Figure F.13, but divides the pre-YCC periods into three subperiods, with different trading hours.

G Simple Theoretical Framework

We lay out a simple theoretical framework that can account for our empirical findings. The framework and the discussion here build heavily on those of Duarte et al. (2025) and Caballero et al. (2024).

G.1 Flexible Interest Rate Environment

Consider a two-period economy, $t = 0, 1$. The economy is populated by a representative household and a consolidated central bank and government. There are two assets: stocks and bonds. Stocks are claims to stochastic output in period 1, Y_1. The supply of stock is normalized to one, and we let P_0 be the price of stocks in period 0. Bonds are risk-free assets in zero net supply with a gross interest rate of R.

We assume that the representative household has log utility over consumption in both periods, C_0 and C_1, with a discount factor of β. The representative household solves the following problem:

$$\begin{aligned} \max_{C_0, C_1, S_0} \quad & \ln C_0 + \beta \mathbb{E}_0 \left[\ln C_1 \right] \\ \text{s.t.} \quad & C_0 + S_0 = W_0 \\ & C_1 = \left[\left(\frac{Y_1}{P_0} - R \right) \alpha(P_0, R) + R \right] S_0, \end{aligned}$$

where $W_0 \equiv P_0 + Y_0$ is the initial wealth of the household, Y_1 / P_0 is the return from the stocks, and $\alpha(P_0, R) > 0$ is the share of stocks in the household's portfolio as a function of the stock price and the interest rate. The expectation is taken over the stochastic output in period 1, Y_1, conditional on information available at $t = 0$.

We do not specify the functional form of $\alpha(P_0, R)$ and keep it an arbitrary function. The frictionless benchmark is nested as a special case with $\alpha(P_0, R) = \arg\max_\alpha \mathbb{E} \left[\ln \left(\alpha \left(\frac{Y_1}{P_0} - R \right) + R \right) \right]$. At the other extreme, the portfolio share could be exogenously fixed, $\alpha(P_0, R) = \bar{\alpha}$. Our flexible specification allows for a continuum of cases between these two extremes.

Under log utility, the optimal consumption and saving decisions are given by:

$$C_0 = \frac{W_0}{1 + \beta} = \frac{P_0 + Y_0}{1 + \beta}. \tag{G.1}$$

and

$$S_0 = \frac{\beta}{1+\beta}(P_0 + Y_0). \tag{G.2}$$

The central bank trades stocks and bonds at $t = 0$, and its budget constraint is given by:

$$P_0 Q^{CB} + B^{CB} = 0 \tag{G.3}$$

where Q^{CB} is the quantity of stocks purchased, and B^{CB} is the quantity of bonds purchased by the central bank. Note that $B^{CB} < 0$ means sales of bonds by the central bank. At $t = 1$, the budget constraint of the consolidated government is given by $G_1 = Y_1 Q^{CB} + R B^{CB}$, where G_1 is the government spending in period 1. The government spending residually adjusts to balance the government budget.

The goods market clearing condition at $t = 0$ is

$$C_0 = Y_0, \tag{G.4}$$

and the stock market clearing condition at $t = 0$ is

$$\alpha(P_0, R) S_0 + P_0 Q^{CB} = P_0. \tag{G.5}$$

G.2 Central Bank Stock Purchase under a Flexible Interest Rate

Now we study the impact of the central bank's stock purchase in the model. We consider an increase in Q^{CB} accompanied by a decrease in B^{CB} to satisfy the budget constraint of the central bank (G.3).

Note that, perhaps surprisingly, it follows from the goods market clearing condition (G.4) together with the consumption function (G.1) that such a policy leaves the stock price entirely unchanged. To see this, substitute (G.1) into (G.4) and solve for P_0 to obtain:

$$P_0 = \beta Y_0, \tag{G.6}$$

which is entirely invariant to the central bank's balance sheet (note Y_0 is exogenous here).

This result does not mean that the central bank stock purchase is neutral. In fact, it is not. Given the knowledge of $P_0 = \beta Y_0$, it follows from (G.2) that $S_0 = \beta Y_0$. Substituting $P_0 = \beta Y_0$ and $S_0 = \beta Y_0$ into the stock market clearing condition (G.5) and totally differentiating,

we have

$$\frac{dR}{dQ^{CB}} = -\frac{1}{\partial \alpha(P_0, R)/\partial R}. \tag{G.7}$$

Therefore, as long as the portfolio share of stocks $\alpha(P_0, R)$ is decreasing in the interest rate R, $\partial \alpha(P_0, R)/\partial R < 0$, the central bank stock purchase raises the interest rate.

We summarize the results as follows, echoing Proposition 1 of Duarte et al. (2025).

Proposition 1. *Suppose that $\partial \alpha(P_0, R)/\partial R < 0$. Then, the central bank stock purchase has no effect on the stock price, but raises the interest rate:*

$$\frac{dP_0}{dQ^{CB}} = 0, \quad \frac{dR}{dQ^{CB}} > 0. \tag{G.8}$$

Note that the assumption of $\partial \alpha(P_0, R)/\partial R < 0$ is extremely natural to impose, and it is satisfied in any standard portfolio choice problem. The intuition is as follows. An inflow into the stock market generally puts upward pressure on the stock price, especially so when the portfolio share is inelastic to the stock price, i.e., $|\partial \alpha(P_0, R)/\partial P_0|$ is low. The higher stock price in turn induces a stock market wealth effect on consumption. However, the aggregate output is fixed, so a higher stock price and higher consumption are incompatible with the goods market clearing condition. This implies that in equilibrium, the stock price cannot rise. If the stock price remains the same, then what would equilibrate the excess demand in the stock market? The answer is the interest rate. The higher interest rate shifts the portfolio away from stocks and thereby clears the stock market.

G.3 Central Bank Stock Purchase with a Fixed Interest Rate

Now we consider the central bank stock purchase with a fixed interest rate. As in the New Keynesian tradition as well as its augmentation with the macro-finance models (Caballero and Simsek, 2020; Caballero et al., 2024), we assume that the central bank fixes the (real) interest rate and output at $t = 0$ is demand-determined to clear the goods market. From (G.6), we have that output is a one-to-one function of the stock price, $Y_0 = P_0/\beta$. Substituting into (G.2), saving is given by $S_0 = P_0$. Substituting $S_0 = P_0$ into the stock market clearing condition (G.5) and totally differentiating, we have

$$\frac{dP_0}{dQ^{CB}} = -\frac{1}{\partial \alpha(P_0, R)/\partial P_0}. \tag{G.9}$$

Therefore, as long as the portfolio share of stocks $\alpha(P_0, R)$ is decreasing in the stock price P_0, $\partial\alpha(P_0, R)/\partial P_0 < 0$, the central bank stock purchase raises the stock price. Consequently, output also increases,

$$\frac{dY_0}{dQ^{CB}} = -\frac{1}{\beta\partial\alpha(P_0, R)/\partial P_0}. \tag{G.10}$$

We summarize the results as follows.

Proposition 2. *Suppose that $\partial\alpha(P_0, R)/\partial P_0 < 0$ and the interest rate is kept fixed by the central bank. Then, the central bank stock purchase raises the stock price and output, while keeping the interest rate unchanged:*

$$\frac{dP_0}{dQ^{CB}} > 0, \quad \frac{dY_0}{dQ^{CB}} > 0, \quad \frac{dR}{dQ^{CB}} = 0. \tag{G.11}$$

When the interest rate is fixed, the excess demand in the stock market is equilibrated by the rise in the stock price. The rise in the stock price induces a stock market wealth effect on consumption, and the goods market clears through a demand-induced increase in output, as in the New Keynesian model.

Propositions 1 and 2 jointly provide a unified explanation for our empirical findings. When the interest rate is flexible, the excess demand for stocks induced by the central bank stock purchase is equilibrated entirely by the rise in the interest rate. This is because any movement in the stock price is incompatible with the goods market clearing condition. When the interest rate is fixed, the excess demand for stocks is equilibrated entirely by the rise in the stock price.

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